

IS–LM/AD–AS Model

ECON202 MACROECONOMICS-1

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1. Start by asking: “How do we analyze the economy when all three markets—labor, goods, and assets—interact simultaneously?” **The IS-LM model was developed by John Hicks in 1937**, just one year after Keynes published the General Theory.
2. This chapter synthesizes previous chapters into a unified framework that economists use for policy analysis. **Fun fact: Hicks won the Nobel Prize in 1972 partly for this contribution**, though he later expressed doubts about the model’s usefulness!
3. Emphasize that this is the workhorse model used by central banks and government agencies worldwide. **The Federal Reserve’s FRB/US model has over 350 equations but its core is still IS-LM logic**. The Bank of England, ECB, and Bank of Japan all use variants.

Roadmap of Lecture

Learning Objectives

The **FE** Line: Equilibrium in the Labor Market

The **IS** Curve: Equilibrium in the Goods Market

The **LM** Curve: Asset Market Equilibrium

General Equilibrium in the Complete **IS–LM** Model

Worked-Out Numerical Exercise for Solving the IS–LM

What Will You Master Today?

- Discuss factors that explain the full-employment (FE) line
- Discuss factors that affect the **IS** curve, which represents equilibrium in the goods market
- Discuss factors that affect the **LM** curve, which represents equilibrium in the asset market
- Describe the conditions necessary for general equilibrium using the **IS-LM** model
- Discuss the role of price adjustment in achieving general equilibrium
- Explain the fundamentals and implications of the AD-AS model

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1. Ask students: "Why do we need three separate curves to analyze the macroeconomy?" **Walras' Law tells us if N-1 markets clear, the Nth market must also clear**—but we still analyze all three for completeness.
2. Each curve represents equilibrium in one market: FE for labor market, **IS** for goods market, **LM** for asset market. **Historical note: Before IS-LM, economists struggled to analyze multiple markets simultaneously**—this was Hicks's breakthrough.
3. By the end of this chapter, students should be able to trace any shock through all three markets. **Real-world application: This is exactly what Fed economists do before every FOMC meeting**—they simulate various scenarios using **IS-LM** logic.
4. General equilibrium means all three markets clear simultaneously—this is the economy's "resting point." **Arrow-Debreu (1954) proved existence of general equilibrium under certain conditions**—both won Nobel Prizes for this work.
5. The debate between Classical and Keynesian economists centers on how quickly prices adjust to restore equilibrium. **Empirical evidence: Menu costs average 7-11% of revenue (Levy et al. 1997)**, explaining price

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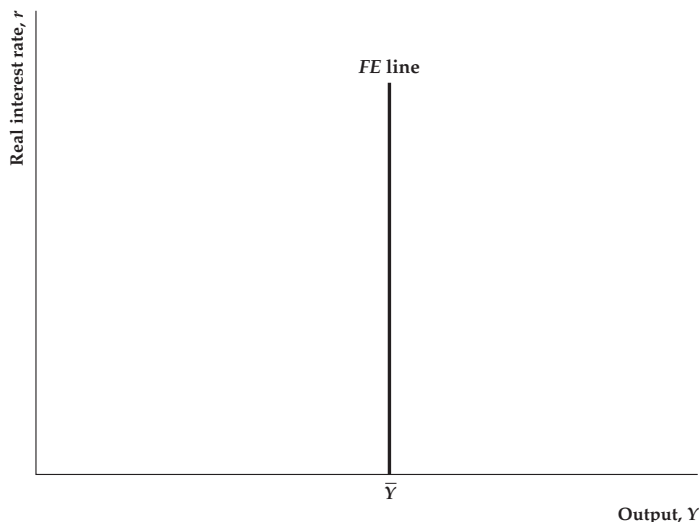
Why Doesn't Interest Rate Affect How Much We Work?

- Labor market in Chapter 3 showed how equilibrium in the labor market leads to employment at its full-employment level N and output at its full-employment level $\bar{Y}$
- If we plot output against the real interest rate, we get a vertical line, since labor market equilibrium is unaffected by changes in the real interest rate

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1. Key term: Full-employment output ($\bar{Y}$) is the level of output produced when labor market is in equilibrium—not necessarily zero unemployment. **In the US, this corresponds to about 4-5% unemployment rate**—the “natural rate” (Friedman 1968).
2. Ask: “Why doesn’t the real interest rate affect labor supply or labor demand?” Answer: Workers and firms care about real wages, not interest rates, when making employment decisions. **Empirical evidence: Intertemporal labor supply elasticity is near zero** (MaCurdy 1981, Altonji 1986).
3. The vertical FE line is a fundamental property: it shows that supply-side factors determine long-run output. **This is Say’s Law: “Supply creates its own demand”** in the long run. Jean-Baptiste Say (1803) was right about the long run!

The Vertical Line That Determines Our Economic Potential



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1. **The full-employment (FE) line represents labor market equilibrium. When the labor market is in equilibrium, employment equals its full-employment level, N , and output equals its full-employment level, $\bar{Y}$, regardless of the value of the real interest rate. Thus the FE line is vertical at $Y = \bar{Y}$.**
2. Point out the vertical nature of the FE line—this is crucial for understanding Classical economics. **The position is determined by the Solow growth model factors: technology, capital, and labor.**
3. This diagram is the basis for the AD–AS model later. **In AD–AS, the vertical long-run aggregate supply (LRAS) curve is equivalent to the FE line.** Both represent the economy's productive capacity.

What Makes an Economy's Productive Capacity Change?

Factors that shift the FE line

- The full employment level of output is determined by the full-employment level of employment and the current levels of capital and productivity
- Any change in these variables shifts the FE line

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1. Recall the production function: $Y = AF(K, N)$ where A is productivity, K is capital, N is employment. **Cobb-Douglas form: $Y = AK^\alpha N^{1-\alpha}$ where $\alpha \approx 0.3$ for most countries** (Gollin 2002).
2. Anything that increases $\bar{N}$, K , or A shifts FE rightward—these are supply-side improvements. **China's FE line shifted right by 10% annually during 1980-2010**—mostly from capital accumulation and technology transfer.
3. Key question: "Can demand-side policies like increasing government spending shift the FE line?" Answer: No, only supply-side factors matter for FE. **Reagan's supply-side economics (1980s) focused on this**—cut taxes to incentivize work and investment, not to boost demand.

When Good Things (and Bad Things) Happen to Productivity

- The full-employment line shifts right because of:
 - A beneficial supply shock
 - $\uparrow$ in labor supply
 - $\uparrow$ in the capital stock
- The full-employment line shifts left when the opposite happens to the three factors above

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1.
 - **Beneficial supply shock** shifts FE right: More output can be produced for the same amount of capital and labor; if MPN rises, labor demand increases
 - **Increase in labor supply** shifts FE right: Equilibrium employment rises, raising full-employment output
 - **Increase in capital stock** shifts FE right: More output can be produced with the same amount of labor; increased capital may increase MPN

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Worked-Out Numerical Exercise for Solving the IS–LM

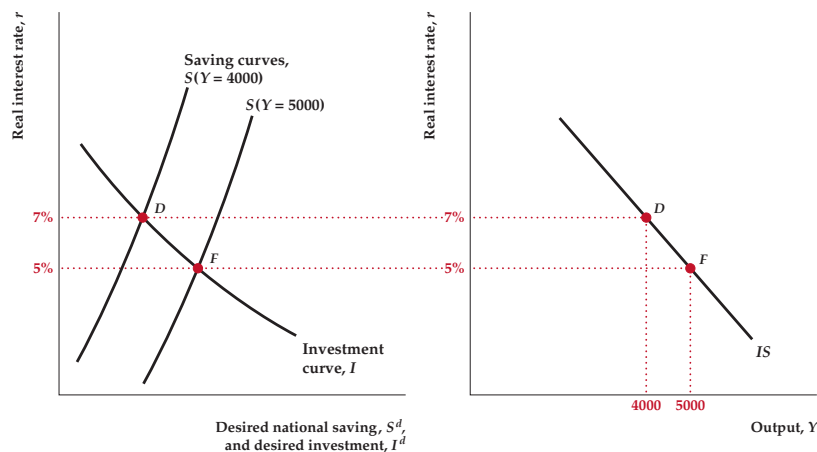
When Does Saving Equal Investment? The Interest Rate Story

- The goods market clears when **desired investment** equals **desired national saving**
- Adjustments in the **real interest rate** bring about equilibrium
- For any level of output Y , the **IS** curve shows the real interest rate r for which the goods market is in equilibrium

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1. Key definition: **IS** stands for Investment-Saving equilibrium. The **IS** curve shows all (Y, r) combinations where $S^d = I^d$. **Wicksell (1898) first described this “natural rate of interest”** that equilibrates saving and investment.
2. The real interest rate is the “price” that equilibrates the goods market—higher r reduces investment and increases saving. **Current US real rate $\approx 2\%$** (nominal 5% minus expected inflation 3%). **Historically averages 2-3% in developed countries.**
3. Opening question: “When consumers save more, what happens to interest rates?” This introduces the paradox of thrift discussion later. **Japan’s excess savings drove rates to zero in 1990s**—the “liquidity trap” Keynes warned about!

Deriving the IS Curve



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1. Top panel shows saving-investment diagram; bottom panel shows resulting **IS curve**. **This two-panel derivation technique was Hicks's innovation**—made complex relationships visual and intuitive.
2. Walk through the derivation: higher Y shifts saving right, intersection moves down, so r falls—hence downward-sloping **IS**. **Empirically, IS slope ≈ -0.5 :** 1% increase in interest rate reduces output by 0.5% (Ramey 2016).
3. This two-panel diagram technique is essential for understanding the model—make sure students grasp the connection. **MIT and Chicago PhD prelims always include IS-LM derivation questions**—master this now!

The IS Curve: Equilibrium in the Goods Market

Key features:

- The saving curve slopes upward because a higher real interest rate increases saving
- $\uparrow$ in output shifts the saving curve to the right
- The investment curve slopes downward because a higher real interest rate reduces the desired capital stock

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1. Why does higher r increase saving? Substitution effect (future consumption cheaper) dominates income effect. **Empirically, savings elasticity to interest rates $\approx 0.2-0.4$ (Boskin 1978). Exception: Very wealthy may have negative elasticity** (income effect dominates).
2. Why does higher Y shift saving right? Higher income means people save more at every interest rate. **Marginal propensity to save ≈ 0.2 in US**, meaning 20 cents of each extra dollar is saved. **In China, MPS ≈ 0.45** —cultural and institutional differences matter!
3. Investment is downward sloping because higher r makes fewer projects profitable—only high-return projects undertaken. **Rule of thumb: 1% rise in rates reduces investment by 2-3% (Caballero 1999). Housing investment even more sensitive**—elasticity of -8!

The IS Curve Logic: Connecting Output, Saving, and Interest Rates

Consider two different levels of output

- At the higher level of output, the saving curve is shifted to the right
- Since the investment curve is downward sloping, equilibrium at the higher level of output has a lower real interest rate
- Thus a higher level of output must lead to a lower real interest rate, so the **IS** curve slopes downward
- The **IS** curve shows the relationship between the real interest rate and output for which investment equals saving

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1. This is the key insight: higher output generates more saving, which pushes interest rates down to stimulate investment. **Bernanke's "global savings glut" hypothesis (2005)**—Asian savings flooded US markets, lowering rates and inflating housing bubble.
2. Intuition: Abundant output means abundant saving, which makes borrowing cheaper. **Germany and Japan run current account surpluses of 7-8% of GDP**—their excess savings depress global interest rates.
3. Ask: "Why don't we see this relationship break down in reality?" Discuss cases where **IS** might be steep or flat. **IS becomes vertical at zero lower bound**—monetary policy loses traction. **IS flattens during credit crunches** when banks won't lend regardless of rates.

Alternative View: How Interest Rates Control Spending

- Beginning at equilibrium, suppose the real interest rate rises
- The increased real interest rate causes people to increase saving and reduce consumption, and causes firms to reduce investment
- So higher real interest rates are associated with lower output
- To restore equilibrium, the quantity of goods supplied would have to decline
- So higher real interest rates are associated with lower output, that is, the **IS** curve slopes downward

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1. This is the aggregate demand interpretation: higher r reduces both C and I , so $Y = C + I + G$ falls. **Consumption elasticity to rates ≈ -0.3 , investment elasticity ≈ -2.0** —investment much more sensitive.
2. This connects to students' intuition about monetary policy: Fed raises rates to "cool down" the economy. **Volcker raised rates to 20% in 1981** to break inflation—caused severe recession but worked! **Each 1% rate hike reduces GDP by 0.5% after 2 years** (Romer and Romer 2004).
3. Both interpretations (saving-investment and aggregate demand) are equivalent—use whichever helps students understand better. **Keynes emphasized aggregate demand; Classical economists prefer saving-investment**—same math, different intuition.

What Shifts IS? The Saving-Investment Balance Story

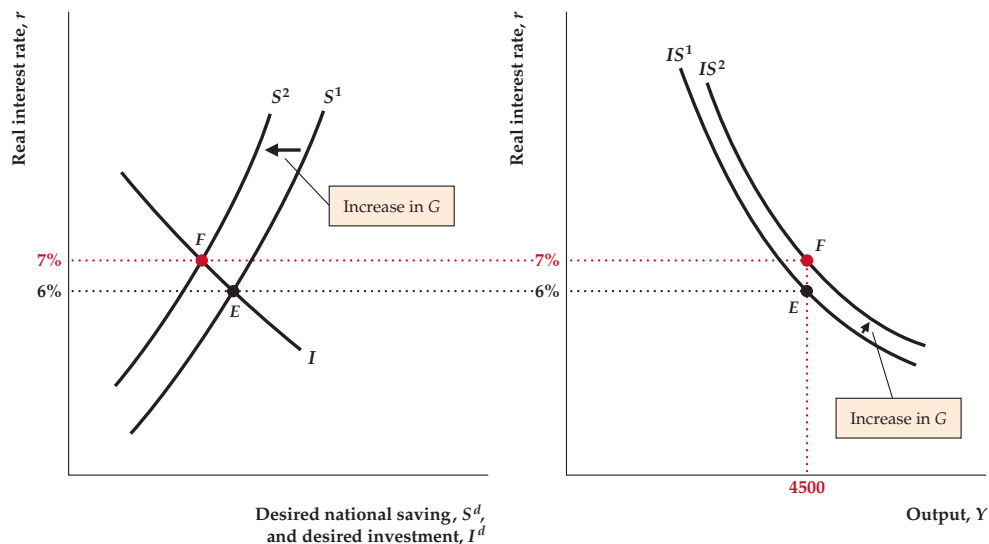
Factors that shift the **IS** curve

- Any change that reduces desired national saving relative to desired investment shifts the **IS** curve up and to the right
- Intuitively, imagine constant output, so a reduction in saving means more investment relative to saving; the interest rate must rise to reduce investment and increase saving
- Example of temporary increase in government purchases

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1. Mnemonic: "Less saving or more investment shifts **IS** right" because interest rate must rise to restore equilibrium. **Animal spirits (Keynes's term) can shift IS dramatically**—dot-com boom shifted **IS** right, crash shifted it left.
2. Example: Tax cuts reduce saving (if no Ricardian equivalence), shifting **IS** right—this is expansionary fiscal policy. **Trump tax cuts (2017) shifted IS right by about 0.7% of GDP**—boosted growth temporarily but increased deficits.
3. Key question: "Does an increase in government spending shift **IS** right or left?" Answer: Right, because it reduces national saving ($S = Y - C - G$). **WW2 spending shifted IS massively right**—deficit reached 27% of GDP! Ended Great Depression but caused inflation.

How Does Government Spending Shift the IS Curve?



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1. Top panel: Government spending reduces national saving (shifts saving left), raising equilibrium r . **Each 1% of GDP increase in G raises rates by 0.3-0.5 percentage points** (Laubach 2009).
2. An alternative way of stating this is that A change that increases aggregate demand for goods shifts the **IS** curve up and to the right
 - 2.1 In this case, the increase in aggregate demand for goods exceeds the supply
 - 2.2 The real interest rate must rise to reduce desired consumption and investment and restore equilibrium
3. A change that increases desired national saving relative to desired investment shifts the **IS** curve down and to the left
4. This is the classic Keynesian case for fiscal stimulus, but note it raises interest rates, potentially "crowding out" private investment. **Crowding out estimated at 30-60 cents per dollar of spending** (Blanchard and Perotti 2002). **In open economy, also crowds out net exports**—twin deficits problem.

Factors that shift the IS due to Increase in

1. expected future output $\Rightarrow$ Up
2. wealth $\Rightarrow$ Up
3. A temporary increase in government purchases $\Rightarrow$ Up
4. in the expected future marginal product of capital $\Rightarrow$ Up
5. taxes (if Ricardian equivalence doesn't hold) $\Rightarrow$ Down
6. A decrease in the effective tax rate on capital $\Rightarrow \Downarrow$

The IS curve shifts $\Downarrow$ and to the left when the opposite happens to the six factors above

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1. **Expected future output** increases: **IS** shifts up and to the right – desired saving falls
2. **Wealth** increases: **IS** shifts up and to the right – desired saving falls
3. **Government purchases** increases: **IS** shifts up and to the right – desired saving falls
4. **Expected future TFP** increases: **IS** shifts up and to the right – desired investment increases
5. **Taxes** increase: No change or **IS** shifts down and to the left – depends on Ricardian equivalence
6. **Effective tax rate on capital** increases: **IS** shifts down and to the left – desired investment falls

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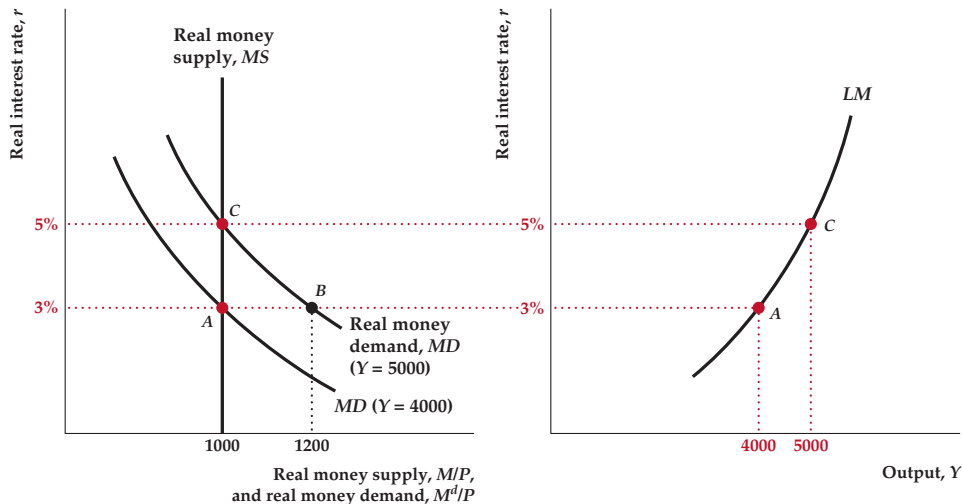
Why Do Bond Prices Fall When Interest Rates Rise?

- The interest rate and the price of a nonmonetary asset
- The price of a nonmonetary asset is inversely related to its interest rate or yield
- Example: A bond pays \$10,000 in one year; its current price is \$9615, and its interest rate is 4%
- If the price falls to \$9524, its yield rises to 5%

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1. A nonmonetary asset like a bond is an IOU - you pay money today to receive more money later. The interest rate (or yield) is your reward for lending. These two values - what you pay (price) and what you earn (yield) - always move in opposite directions.
2. This inverse relationship is mathematical, not behavioral. Since bonds promise fixed future payments, only today's price can adjust. If a bond pays \$1000 next year and costs \$950 today, $\text{yield} = 5.3\%$. If price drops to \$900, yield jumps to 11.1%.
3. Our example: Pay \$9,615 today, receive \$10,000 in one year = \$385 profit = 4% yield. If market rates rise to 5%, this bond's price must fall to \$9,524 (giving \$476 profit) to stay competitive. Markets self-adjust through supply and demand.
4. Real impact: When the Fed raises rates, existing bondholders lose money as their bonds' prices fall to match new higher yields. Conversely, falling rates benefit current bondholders. This principle explains why "safe" bond investments can still lose value.

Building the LM Curve



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- By what mechanism is equilibrium restored?
 - Starting at equilibrium, suppose output rises, so real M^d increases
 - The rise in people's demand for money makes them sell nonmonetary assets
 - The price of those assets falls and the real interest rate rises
- This is the portfolio balance mechanism: When people need more money, they sell bonds, driving bond prices down and interest rates up. **Happens in microseconds in modern markets**—algorithmic trading ensures instant adjustment.
- Key insight: The interest rate adjusts to make people willing to hold the existing M^s . **Like a bidding war**: If everyone wants more cash, they must offer higher returns on alternatives. **Money market funds now pay 5%**—competing for deposits.
- Ask: "Why can't people just 'create' more money when they need it?"
 Answer: Money supply is controlled by the central bank. **Only Fed can create reserves; banks can create deposits but constrained by reserve requirements. Crypto tried to solve this—but isn't real money (yet?).**

Why Does the LM Curve Slope Upward?

- The **LM** curve shows the combinations of the real interest rate and output that clear the asset market
- For any given level of output, the **LM** curve shows the real interest rate necessary to equate real M^d and supply
- Thus the **LM** curve slopes upward from left to right

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1. **LM's** upward slope contrasts with **IS's** downward slope—this creates a unique equilibrium intersection. **Mathematically, need opposing slopes for unique solution**—if both sloped same way, might have no or infinite equilibria.
2. Intuition: Economic expansion (higher Y) increases demand for money, tightening financial conditions (higher r). **This is automatic monetary tightening**—economy has built-in stabilizer. **Problem: Sometimes works too well**, choking off recoveries.
3. The **LM** curve represents monetary policy's "passive" response—what happens if Fed doesn't change M^s . **Modern Fed uses interest rate targets instead**—effectively makes **LM** horizontal at target rate. This is the "Taylor Principle" in action.

What Makes the LM Curve Shift?

- Any change that reduces real M^s relative to real M^d shifts the **LM** curve up
 - For a given level of output, the reduction in real M^s relative to real M^d causes the equilibrium real interest rate to rise
 - The rise in the real interest rate is shown as an upward shift of the **LM** curve
- A change that increases real M^s relative to real M^d shifts the **LM** curve down and to the right

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1. Mnemonic: "More money shifts **LM** down" because lower interest rates are needed to get people to hold extra money. **QE1 (2008): Fed created \$1.75 trillion—massive LM shift down. QE2: another \$600 billion. QE3: \$85 billion/month!**
2. Real M^s is M/P —can change through nominal M^s M or price level P . **During hyperinflation, LM shifts up continuously** as P explodes. **Weimar Germany 1923:** Prices doubled every 3 days—**LM** shifting up hourly!
3. Key question: "If the Fed increases M by 10% but prices also rise 10%, does **LM** shift?" Answer: No, real M^s unchanged. **This is why Fed targets inflation at 2%—keeps real variables stable. Money illusion fools people temporarily but not forever.**

The LM curve shifts down and to the right because of:

- ↑ in the nominal M^s
- ↑ in the price level
- ↑ in expected inflation
- ↑ in the nominal interest rate on money
- ↑ in Other Factors that ↑ the real money demand
 - ↑ in wealth
 - ↑ in the risk of alternative assets
 - ↓ in the liquidity of alternative assets
 - ↓ in the efficiency of payment technologies

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1. First four factors are most important. Expected inflation shifts **LM** down because it reduces real return on money. **Fisher effect:** Nominal rate = Real rate + Expected inflation. **If people expect 5% inflation, holding cash means 5% real loss**—they flee to other assets.
2. Modern example: Mobile payment apps (Venmo, PayPal) increase payment efficiency, reducing M^d and shifting **LM** down. **China's mobile payments reached 83% of GDP**—dramatically reduced cash demand. **Sweden aLMost cashless**—only 1% of transactions in cash. **Kenya's M-PESA revolutionized monetary system.**
3. Students often confuse nominal vs. real effects—emphasize that **LM** depends on real M^s M/P . **Central banks can control M but not P directly**—this is the heart of monetary policy challenges. **Venezuela printed money but prices rose faster**—**LM** actually shifted left!

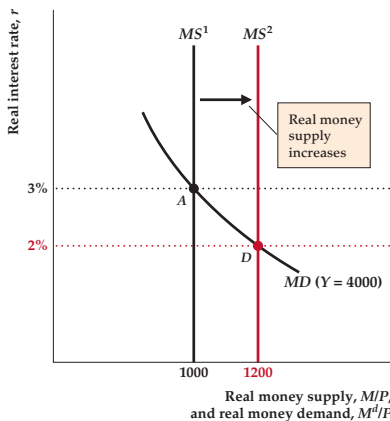
Quick Guide: Four Key LM Shifters to Remember

- **Nominal M^s** increases: **LM** shifts down and to the right – real M^s increases
- **Price level** increases: **LM** shifts up and to the left – real M^s falls
- **Expected inflation** increases: **LM** shifts down and to the right – demand for money falls
- **Nominal interest rate on money** increases: **LM** shifts up and to the left – demand for money increases

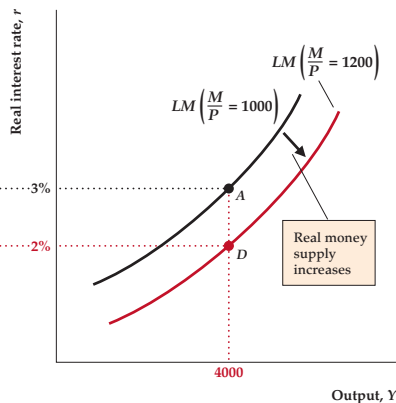
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1. First factor is expansionary monetary policy; second shows how inflation tightens monetary conditions. **Fed's balance sheet grew from \$900 billion (2007) to \$9 trillion (2022)**—unprecedented peacetime monetary expansion. **Now doing QT**—shrinking by \$95 billion/month.
2. Expected inflation effect: If people expect 5% inflation, holding cash loses 5% value, so they hold less money. **1970s Great Inflation**: Expected inflation hit 10%— M^d collapsed, velocity soared. **Volcker had to raise rates to 20%** to restore M^d .
3. Some banks now pay interest on checking accounts—this increases M^d , shifting **LM** up. **Online banks offer 4-5% on checking**—blurs distinction between money and bonds. **Fed now pays 5.4% on reserves**—controversial policy tool. Makes traditional **LM** analysis trickier.

Shifts the LM Curve : How Money Creation Affects Markets



(a)



(b)

- Changes in the real M^s
- $\uparrow$ in the real M^s shifts the **LM** curve down and to the right

1. Top panel: Money supply curve shifts right, lowering equilibrium interest rate at each Y . **Modern twist:** Fed now adjusts interest on reserves rather than money quantity—same effect, different mechanism.
2. Bottom panel: **LM** curve shifts right—economy can support higher Y at same r , or same Y at lower r . **Each 1% increase in M/P lowers rates by 0.05-0.1 percentage points** (Lucas 2000). Effect has weakened as financial innovation progressed.
3. This is expansionary monetary policy graphically—memorize this diagram for policy analysis. **Central banking is 90% shifting this curve around**—rest is communication and psychology. **"Fed speak"** is designed to shift **expectations**, which shifts **LM** without changing M .

When Money Supply and Prices Move at Different Speeds

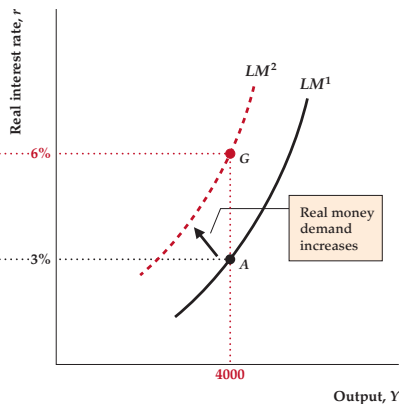
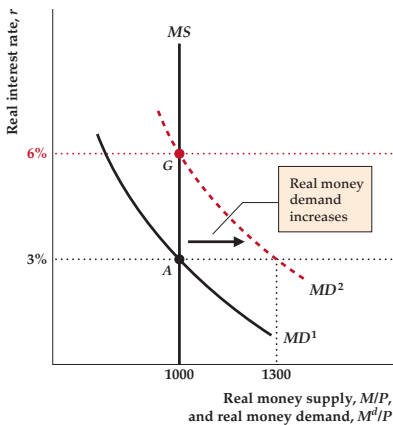
Changes in the real M^s

- A drop in real M^s shifts the **LM** curve up and to the left
- The real M^s changes when the nominal M^s changes at a different rate than the price level

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1. This is contractionary monetary policy—Fed did this in early 1980s under Paul Volcker to fight inflation. **Volcker's Saturday Night Special (Oct 6, 1979)**—changed operating procedures, sent rates soaring. **Caused 1981-82 recession**—unemployment hit 10.8%. But it worked—**inflation fell from 14% to 3%**.
2. Key insight: If M grows 3% but P grows 5%, real M^s falls—this is stealth monetary tightening. **Turkey's current problem**: Money growing 30% but inflation 60%—**LM** shifting left despite money printing! **Argentina similar**—can't print way out of inflation.
3. Deflation (falling P) increases real M^s even if Fed does nothing—happened in Great Depression. **1929-1933: Prices fell 25%**, real M^s rose despite bank failures. **But didn't help**—liquidity trap plus debt deflation (Fisher 1933). **Japan's mild deflation 1990s-2010s** kept shifting **LM** right—couldn't generate inflation despite trying.

Shifts the LM Curve : An Increase in Real Money Demand



- $\uparrow$ in real M^d shifts the **LM** curve up and to the left
- Similarly, A drop in real M^d shifts the **LM** curve down and to the right

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1. Top panel: Money demand shifts right, raising equilibrium r at each Y . **Banking crises cause this**—everyone wants liquidity simultaneously. **Lehman Brothers collapse**: Money market funds “broke the buck,” panic ensued.
2. Bottom panel: **LM** shifts left—contractionary effect without Fed action. **Fed’s job during crisis**: Offset private sector’s increased M^d with increased supply. **“Lender of last resort”** function—Bagehot (1873) defined this role.
3. This happened during financial crises: Panic increases M^d , tightening financial conditions automatically. **1907 Panic**: J.P. Morgan personally acted as central bank. Led to Fed creation 1913. **2008: Fed learned lesson**—flooded system with liquidity. **2020: Even faster response**—prevented financial collapse despite economic shutdown.

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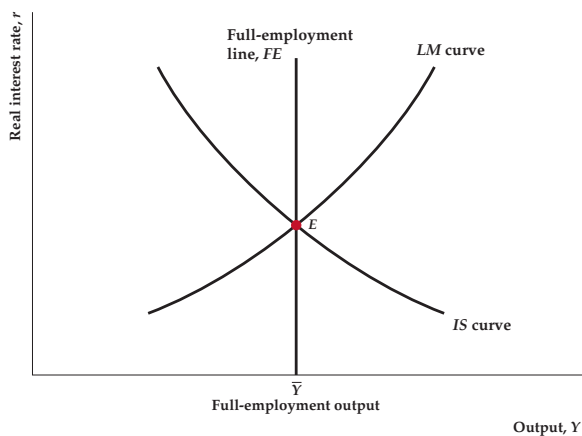
General Equilibrium in the Complete IS–LM Model

- When all markets are simultaneously in equilibrium there is a **general equilibrium**
- This occurs where the FE, **IS**, and **LM** curves intersect

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1. General equilibrium means three things: Full employment (FE), saving equals investment (**IS**), and M^s equals M^d (**LM**). **Walras (1874) first conceived this**—took 80 years to prove existence (Arrow-Debreu 1954). **Both won Nobels** for this fundamental contribution.
2. This is the economy's "natural resting point" where no market has excess supply or demand. **Like ball in bowl**—always returns to bottom. **Question: How fast?** Classicals say quickly, Keynesians say slowly. **Reality: 6-18 months typically** (Romer 2012).
3. Ask: "What happens if the three curves don't intersect at one point?" Answer: Price level adjusts to restore equilibrium. **This is Say's Law in action**—supply creates its own demand through price adjustment. **Works eventually but "eventually" might be years**—hence role for policy.

The key Diagram: All Markets in Perfect Balance



- When all markets are simultaneously in equilibrium $\Rightarrow$ **general equilibrium**
- This occurs where the FE, **IS**, and **LM** curves intersect

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1. This is the master diagram—all three curves intersecting at point E. **Draw this 100 times until automatic**—foundation of macro analysis. Every central bank economist can draw this blindfolded.
2. Point E determines equilibrium output Y_1 and real interest rate r_1 . From here we can derive all other variables. **Employment from production function, consumption from consumption function**, etc. It's a complete system—**six equations, six unknowns**.
3. Students should practice drawing this from memory—it's the foundation of macroeconomic analysis. **PhD qualifying exams love IS-LM questions**. "Explain effects of fiscal expansion when at full employment"—classic question requiring this framework.

Applying the IS–LM framework: A temporary adverse supply shock

- Suppose the productivity parameter in the production function falls temporarily
- The supply shock reduces the marginal productivity of labor, hence labor demand
- Lower employment and lower productivity both reduce the equilibrium level of output, shifting the FE line to the left

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1. Example: Natural disaster, oil price spike, pandemic disrupting production—all reduce productivity temporarily. **Japan tsunami (2011):** Destroyed supply chains, cut productivity 15%. **Texas freeze (2021):** Knocked out refineries, chemical plants. **Suez Canal blockage (2021):** Reduced global shipping productivity 10%.
2. MPN (marginal product of labor) falls, so firms hire fewer workers at any wage—labor demand curve shifts left. **COVID-19 reduced productivity 5% initially**—social distancing, supply disruptions. **Remote work reduced productivity 10-20%** in some sectors (Barrero et al. 2021).
3. This is a pure supply-side shock—it doesn't affect saving behavior or M^d directly. **Different from demand shock**—affects productive capacity, not spending desires. **Harder for policy to address**—can't print oil or create productivity. Supply shocks are central banker's nightmare.

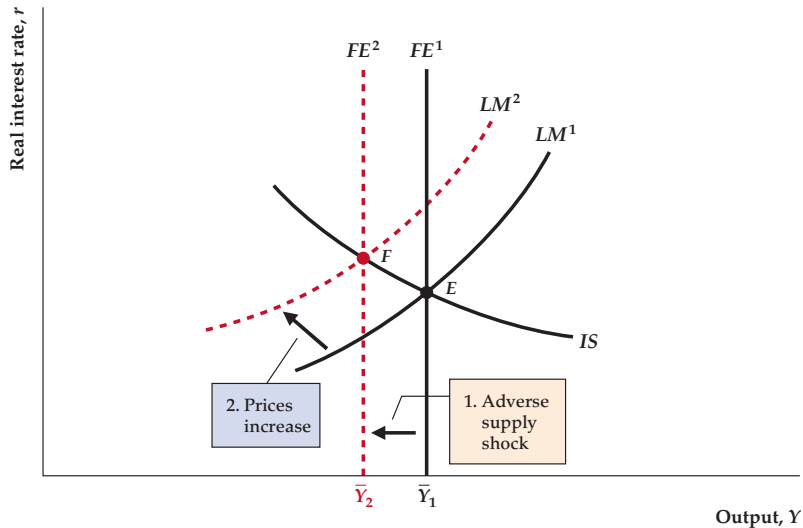
How Does the Economy Self-Correct? The Price Adjustment Story

- There is no effect of a temporary supply shock on the **IS** or **LM** curves
- Since the FE, **IS**, and **LM** curves don't intersect, the price level adjusts
- The price level rises to shift the **LM** curve up and to the left to restore equilibrium

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1. Key insight: Only FE moves, creating disequilibrium. How does equilibrium restore? **Not through government action but through market forces.** This is the self-correcting mechanism economists trust (perhaps too much).
2. Answer: Prices rise. Lower supply with unchanged demand means higher prices (basic supply-demand). **OPEC 1973:** Oil from \$3 to \$12, general price level up 11%. **OPEC 1979:** Oil to \$39, inflation hit 14%. **Ukraine war 2022:** Wheat prices doubled, energy tripled—inflation everywhere.
3. Rising P reduces M/P , shifting **LM** left until it intersects FE and **IS** at the new equilibrium. **This is stagflation mechanism**—falling output plus rising prices. **1970s nightmare scenario:** Recession and inflation simultaneously. **Broke Phillips curve relationship**—economists confused for years.

Applying the IS-LM framework: Figure 9.8



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- Suppose the productivity parameter in the production function falls temporarily
- The supply shock reduces the marginal productivity of labor, hence labor demand
- Lower employment and lower productivity both reduce the equilibrium level of output, shifting the FE line to the left
- There is no effect of a temporary supply shock on the **IS** or **LM** curves
- Since the FE, **IS**, and **LM** curves don't intersect, the price level adjusts
- The price level rises to shift the **LM** curve up and to the left to restore equilibrium

Applying the IS–LM framework: A temporary adverse supply shock

- The inflation rate rises temporarily, not permanently
- Summary: The real wage, employment, and output decline
- The real interest rate and price level are higher
 - There is a temporary burst of inflation as the price level moves to a higher level
 - Since the real interest rate is higher and output is lower, consumption and investment must be lower

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1. Complete list of effects: Y down, N down, w/P down, r up, P up, C down, I down. **Everything bad happens at once**—this is why supply shocks are so damaging. **Living standards fall immediately**—no cushion from borrowing (rates too high).
2. Higher r “crowds out” C and I —this is how the economy adjusts to lower potential output. **Cannot consume what isn’t produced**—budget constraint binds. **1970s: Investment share of GDP fell from 18% to 14%**. Took decade to recover.
3. Ask: “Can the government do anything to help?” Discuss limits of demand-side policy when problem is supply-side. **Nixon tried price controls**—disaster, shortages everywhere. **Ford’s “Whip Inflation Now” buttons**—laughable. **Only Volcker’s harsh medicine worked**—but caused severe recession. Sometimes no good options exist.

The effects of a monetary expansion?

- $\uparrow$ in M^s shifts the **LM** curve down and to the right
- Financial markets respond most quickly;
 1. The FE line is slow to respond, because job matching and wage renegotiation take time
 2. we assume labor market is temporarily out of equilibrium
 3. Short-run equilibrium at the intersection of the **IS** and **LM** curves

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1. This is the classic monetary policy experiment: Fed increases M . What happens? **Friedman's helicopter drop thought experiment**—drop money from sky, what occurs? **Modern version: Fed credits bank reserves electronically**—less dramatic but same effect.
2. Short run vs. long run distinction is crucial—prices are sticky in short run, flexible in long run. **Menu costs, wage contracts, fairness norms** all create stickiness. **Median price changes every 4.3 months** (Nakamura-Steinsson 2008). **But some prices change daily** (gasoline, stocks, commodities).
3. Ask: "Why do financial markets respond faster than labor markets?"
 Answer: Asset prices adjust instantly; wages adjust slowly. **Stock market reacts in milliseconds to Fed announcements. Wages adjust annually at best**—multi-year union contracts common. This lag creates monetary non-neutrality.

The effects of a monetary expansion

- The increase in M^s causes people to try to get rid of excess money balances by buying assets
- This drives the real interest rate down.
- This happens due to inverse relationship between interest rates and asset prices.
- The decline in the real interest rate causes consumption and investment to increase temporarily
- Output is assumed to increase temporarily to meet the extra demand

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1. The Hot Potato Effect: Nobody Wants Extra Money
2. This is the Keynesian transmission mechanism: M up $\rightarrow r$ down $\rightarrow C$ and I up $\rightarrow Y$ up. **Each link can break:** Liquidity trap (rates can't fall), investment pessimism (won't invest despite low rates), etc. **Japan 1990s:** All links broken—"pushing on string."
3. "Hot potato" effect: People don't want to hold extra money, so they buy bonds, driving bond prices up and interest rates down. **Portfolio rebalancing channel**—from money to bonds to stocks to real estate. **Creates asset bubbles if overdone**—Fed's dilemma since 2008.
4. Key assumption: Firms accommodate extra demand by producing more (prices haven't adjusted yet). **Amazon can scale instantly, restaurants can't**—heterogeneous responses. **Supply chains matter**—2021 showed limits when ports clogged, chips scarce. **Can't print semiconductors or dock workers!**

The effects of a monetary expansion

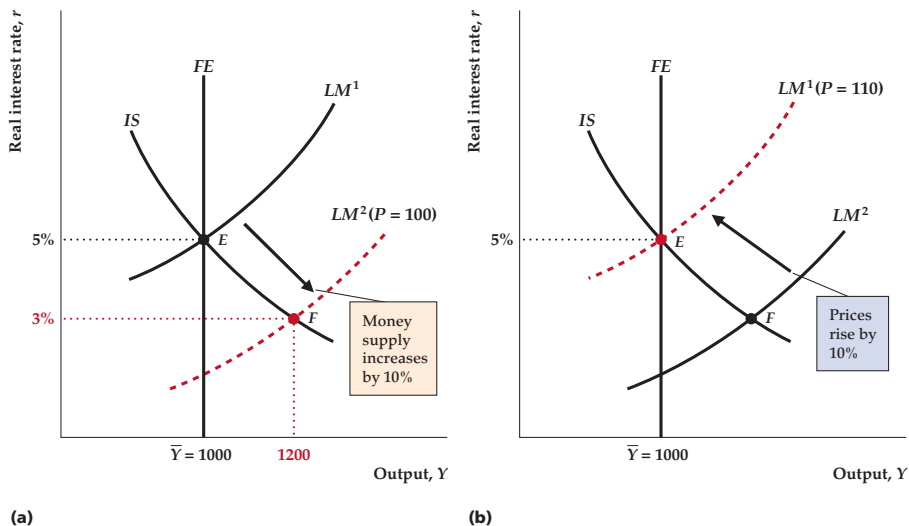
The adjustment of the price level

- Since the demand for goods exceeds firms' desired supply of goods, firms raise prices
- The rise in the price level causes the **LM** curve to shift up
- The price level continues to rise until the **LM** curve intersects with the FE line and the **IS** curve at general equilibrium

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1. This is the self-correcting mechanism: Excess demand triggers inflation, which reverses the initial **LM** shift. **Like compressed spring**—returns to natural position. **But how quickly?** That's the trillion-dollar question.
2. Rising P reduces M/P , shifting **LM** back left until output returns to $\bar{Y}$. **Mathematical certainty if prices flexible**—but big IF. **Downward price rigidity means adjustment asymmetric**—prices rise easily, fall slowly.
3. How long does this take? Classical view: Months. Keynesian view: Years. This is the key debate! **Empirical evidence mixed**: Sometimes fast (small open economies), sometimes slow (Japan 1990s). **Depends on institutions**—flexible economies adjust faster. **US relatively flexible, Europe less so, Japan least**—explains different recession experiences.

Figure 9.9: Effects of a Monetary Expansion



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1. - $\uparrow$ in M^s shifts the **LM** curve down and to the right
 - Financial markets respond most quickly;
 - 1.1 The FE line is slow to respond, matching and wage renegotiation take time
 - 1.2 we assume labor market is temporarily out of equilibrium
 - 1.3 Short-run equilibrium at the intersection of the **IS** and **LM** curves
2. - The increase in M^s causes people to try to get rid of excess money balances by buying assets
 - This drives the real interest rate down
 - The decline in the real interest rate causes consumption and investment to increase temporarily
 - Output is assumed to increase temporarily to meet the extra demand
3. - Since the demand for goods exceeds firms' desired supply of goods, firms raise prices
 - The rise in the price level causes the **LM** curve to shift up
 - The price level continues to rise until the **LM** curve intersects with the FE line and the **IS** curve at general equilibrium

Monetary Neutrality: Why You Can't Print Your Way to Prosperity

- The result is no change in employment, output, or the real interest rate
- The price level is higher by the same proportion as the increase in the M^s
- All real variables (including the real wage) are unchanged
- Nominal values (including the nominal wage) have risen proportionately with the change in M^s

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1. This is monetary neutrality: In long run, money only affects nominal variables (P , W), not real variables (Y , r , w/P). **Classical dichotomy holds eventually**—real and nominal spheres separate. **But Keynes: "In long run we're all dead"**—short run matters for human welfare.
2. If M doubles, P doubles, but M/P is unchanged—real M^s back to original level. **Quantity Theory of Money:** $MV = PY$. If V and Y constant, M and P move proportionally. **Empirically true in long run** (McCandless-Weber 1995)—correlation 0.95 across countries.
3. Classical dichotomy: Real economy (determined by real factors) is independent of monetary factors in long run. **Technology, preferences, and endowments determine real outcomes**—not pieces of paper. **Money is a "veil"** over real economy—important for transactions but not production. **Bitcoin enthusiasts miss this**—changing monetary system doesn't change real productivity.

Why Do All Countries Have Inflation?

- Trend money growth and inflation
- This analysis also handles the case in which the M^s is growing continuously
- If both the M^s and price level rise by the same proportion, there is no change in the real M^s

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1. In reality, M grows continuously (say 5% per year). In steady state, P also grows 5% per year. **This is why all countries have positive inflation— M^s always growing to accommodate economic growth. 2% inflation target is compromise—high enough to avoid deflation, low enough to preserve money's value.**
2. This explains why inflation is always “a monetary phenomenon” (Milton Friedman)—caused by money growth. **Cross-country evidence overwhelms: High money growth = high inflation. Zimbabwe printed money, got hyperinflation. Switzerland disciplined, has low inflation.** No exceptions to this rule over long periods.
3. Ask: “Why do central banks allow money to grow rather than keeping it constant?” Answer: Accommodate economic growth and prevent deflation. **If economy grows 3% but M fixed, P must fall 3%—deflation! Deflation is toxic—raises real debt burdens, causes hoarding. Great Depression lesson: Never allow deflation—err on side of mild inflation.**

Technical Note: Levels vs. Growth Rates

- Often we discuss things in relative terms
- When we talk about “an increase in the M^s ,” we have in mind an increase in the growth rate relative to the trend
- A result that the price level declines can be interpreted as the price level declining relative to a trend

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1. Important technical point: “Increase in M ” often means increase in *growth rate* of M , not level. **Fed typically changes growth rates, not levels.** QE was exceptional—actually changed level dramatically. Normal policy is adjusting growth rates by 0.25-0.5 percentage points.
2. Example: If trend money growth is 3% and Fed increases it to 5%, that’s an “expansionary shock.” **Markets obsess over these small changes**—0.25% move causes billions in trades. **Fed dot plots show expected rate path**—most important financial document globally.
3. Similarly, “falling prices” might mean inflation decreases from 4% to 2%, not actual deflation. **Disinflation vs. deflation**—very different! Disinflation is falling inflation rate (good if from high level). **Deflation is negative inflation** (aLMost always bad). **Japan had deflation, US just disinflation**—explains different policy responses.

The Great Debate: How Fast Do Markets Self-Correct?

- Classical versus Keynesian versions of the **IS-LM** model
- Two key questions in the debate:
 - How rapidly does the economy reach general equilibrium?
 - What are the effects of monetary policy on the economy?

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1. This is the fundamental divide in macroeconomics—not just theoretical, but has major policy implications. **Every Fed chair must decide:** Am I Classical or Keynesian? **Greenspan leaned Classical, Bernanke Keynesian, Powell pragmatic**—shifts with circumstances.
2. Classical answer: Quickly (months). Keynesian answer: Slowly (years). Evidence is mixed. **Small shocks adjust fast, large shocks slowly**—nonlinear dynamics. **Financial crises cause long adjustments**—balance sheet recessions different. **Normal recessions: 1-2 years. Financial crises: 5-7 years** (Reinhart-Rogoff 2009).
3. Your view on this determines whether you support active stabilization policy or laissez-faire. **Republicans typically Classical, Democrats Keynesian**—but exceptions during crises. **Nixon: “We’re all Keynesians now”** (1971). **Trump surprisingly Keynesian**—massive deficits. Biden too. **Academic economists more nuanced**—depends on situation.

Can Markets Heal Themselves?

- Price adjustment and the self-correcting economy
- The economy is brought into general equilibrium by adjustment of the price level
- The speed at which this adjustment occurs is much debated

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1. "Self-correcting" means automatic return to full employment without government intervention. **Like immune system**—body heals itself. **But sometimes needs medicine** (policy). Question is when to intervene vs. let nature take course.
2. If prices adjust quickly, no need for active fiscal/monetary policy—markets heal themselves. **Creative destruction (Schumpeter)**—recessions cleanse economy of inefficiencies. **Zombie firms kept alive by low rates**—Japan's problem. Sometimes pain necessary for health.
3. If prices adjust slowly, case for counter-cyclical policy is strong—don't wait years for adjustment. **Human suffering matters**—unemployment isn't just statistics. **Suicide rates rise 1% for each 1% unemployment increase. Lost generation effects**—youth unemployment scars permanently. Can't ignore these costs.

The Classical View: Markets Clear Quickly

- Classical economists see rapid adjustment of the price level
- So the economy returns quickly to full employment after a shock
- If firms change prices instead of output in response to a change in demand, the adjustment process is **aL**Most immediate

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1. Classical view based on flexible prices: Firms like WalMart change prices daily via computer systems. **Amazon changes prices every 10 minutes**—ultimate flexibility. **Uber surge pricing**—instant market clearing. **Stock market**—continuous price adjustment. Why not all markets?
2. If prices are perfectly flexible, no such thing as involuntary unemployment—labor market always clears. **Any unemployment is voluntary**—workers choosing leisure over work at prevailing wage. **Controversial but logically consistent**. Real unemployment is search/matching process.
3. Policy implication: Don't use activist monetary/fiscal policy—it only creates inflation without helping employment. **Focus on supply side**—education, infrastructure, innovation. **Reagan Revolution based on this. "Government is problem, not solution"**—let markets work. Still influential in policy circles.

Monetary neutrality: Classical vs. Keynesian Views

- The classical view: monetary expansion affects prices quickly with at most a transitory effect on real variables
- Keynesians: a monetary expansion increases output and employment and causes the real interest rate to fall
- Keynesians believe in monetary neutrality in the long run but not the short run

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1. Both sides accept long-run neutrality—debate is about short run (how long is “short run”?). **Friedman: Long and variable lags**—6-18 months typically. **Lucas critique:** Policy changes alter behavior—can’t assume stable relationships. Models break down when regime changes.
2. Empirical evidence: Money shocks affect output for 6-18 months, supporting intermediate view. **Romer-Romer (2004):** Identified monetary shocks have real effects for 2-3 years. **But getting weaker over time**—financial innovation reducing monetary transmission. **Crypto potentially breaks link entirely**—central banks losing control?
3. Conclude: “The truth is probably somewhere in between—adjustment takes months to a few years, not days or decades.” **Synthesis view now dominant**—short-run Keynesian, long-run Classical. **New Keynesian models incorporate both**—sticky prices in DSGE framework. **Policy follows state-contingent rule**—aggressive during crises, neutral otherwise. Pragmatism beats dogma.

Roadmap of Lecture

Learning Objectives

The FE Line: Equilibrium in the Labor Market

The **IS** Curve: Equilibrium in the Goods Market

The **LM** Curve: Asset Market Equilibrium

General Equilibrium in the Complete **IS–LM** Model

Worked-Out Numerical Exercise for Solving the IS–LM

Consider an Economy

Consider an economy that is described by the following equations:

$$C^d = 300 + 0.75(Y - T) - 300r$$

$$T = 100 + 0.2Y$$

$$I^d = 200 - 200r$$

$$L = 0.5Y - 500i$$

$$\bar{Y} = 2500; \quad G = 600; \quad M = 133,200;$$

$$\pi^e = 0.05; \quad P_{sr} = 120$$

In the short run, the price level is fixed at P_{sr} .

Find the short-run and long-run equilibrium values of Y , P , r , C , I , and i .

To help you work through the algebra needed for numerical problems in this chapter, here is a worked-out numerical exercise as an example for solving the IS–LM model:

Step 1: IS Curve Derivation (cont'd)

- Simplify the equation:

- $Y = 300 + 0.75Y - 75 - 0.15Y - 300r + 200 - 200r + 600$

- $0.4Y = 1025 - 500r$

- Solve for r in terms of Y :

- $500r = 1025 - 0.4Y$

- $r = 2.05 - 0.0008Y$ (IS Curve)

The negative slope of the IS curve reflects the inverse relationship between interest rates and output through investment and consumption channels.

Step 2a: General LM Curve

- Asset market equilibrium: $M/P = L$
- Real money demand:
 - $L = 0.5Y - 500i = 0.5Y - 500(r + \pi^e)$
 - $L = 0.5Y - 500(r + 0.05)$
- Real money supply: $M/P = 133,200/P$
- Equilibrium condition:
 - $133,200/P = 0.5Y - 500(r + 0.05)$
 - $500r = 0.5Y - 25 - 133,200/P$

The LM curve represents equilibrium in the money market. The price level P determines the position of the curve.

Step 2b: Short-Run LM Curve

- General LM equation: $r = 0.001Y - 0.05 - 266.4/P$
- In short run, $P = P_{sr} = 120$
- Substitute $P = 120$:
 - $r = 0.001Y - 0.05 - 266.4/120$
 - $r = 0.001Y - 0.05 - 2.22$
 - $r = 0.001Y - 2.27$ (Short-Run LM Curve)

With fixed prices in the short run, the LM curve has a fixed position. The positive slope reflects higher money demand at higher income levels.

Step 3a: Short-Run Equilibrium Values

- Set $IS = LM$ to find equilibrium:
 - $2.05 - 0.0008Y = 0.001Y - 2.27$
 - $4.32 = 0.0018Y$
 - $Y = 2400$ (Short-run output)
- Find interest rate using either curve:
 - IS: $r = 2.05 - 0.0008(2400) = 0.13$
 - LM: $r = 0.001(2400) - 2.27 = 0.13$
 - $r = 0.13$ (Short-run interest rate)

The intersection of IS and LM curves determines the unique short-run equilibrium. Both curves yield the same interest rate, confirming our solution.

Step 3b: Other Short-Run Variables

- Tax revenue: $T = 100 + 0.2(2400) = 580$
- Consumption:
 - $C = 300 + 0.75(2400 - 580) - 300(0.13)$
 - $C = 300 + 1365 - 39 = 1626$
- Investment: $I = 200 - 200(0.13) = 174$
- Nominal interest rate: $i = r + \pi^e = 0.13 + 0.05 = 0.18$
- Verification: $C + I + G = 1626 + 174 + 600 = 2400 = Y \checkmark$

All components of aggregate demand sum to output, confirming internal consistency of our solution.

Step 4a: Long-Run Real Interest Rate

- In long run: $Y = \bar{Y} = 2500$
- Use IS curve to find long-run r :
 - $r = 2.05 - 0.0008(2500)$
 - $r = 2.05 - 2.00$
 - $r = 0.05$ (Long-run interest rate)
- Note: $r = 0.05 = \pi^e$ (Fisher effect at work)

In long-run equilibrium, output returns to potential. The real interest rate adjusts to clear the goods market at full employment.

Step 4b: Long-Run Components

- Tax revenue: $T = 100 + 0.2(2500) = 600$
- Consumption:
 - $C = 300 + 0.75(2500 - 600) - 300(0.05)$
 - $C = 300 + 1425 - 15 = 1710$
- Investment: $I = 200 - 200(0.05) = 190$
- Nominal interest rate: $i = 0.05 + 0.05 = 0.10$
- Verification: $C + I + G = 1710 + 190 + 600 = 2500 \checkmark$

Lower interest rates in long run stimulate both consumption and investment compared to short run.

Step 4c: Long-Run Price Level

- Find money demand at long-run equilibrium:

- $L = 0.5Y - 500i$

- $L = 0.5(2500) - 500(0.10)$

- $L = 1250 - 50 = 1200$

- Set money supply = money demand:

- $M/P = L$

- $133,200/P = 1200$

- $P = 133,200/1200 = 111$ (Long-run price level)

Prices fall from 120 to 111 in moving from short to long run. This increases real money supply, shifting LM curve right.

Step 5: Deriving the AD Curve

- Combine IS and general LM curves:
 - IS: $r = 2.05 - 0.0008Y$
 - LM: $r = 0.001Y - 0.05 - 266.4/P$
- Set IS = LM:
 - $2.05 - 0.0008Y = 0.001Y - 0.05 - 266.4/P$
 - $0.0018Y = 2.10 + 266.4/P$
- Solve for Y:
 - $Y = 1166\frac{2}{3} + 148,000/P$ (AD Curve)

The AD curve shows combinations of P and Y consistent with simultaneous equilibrium in goods and money markets. It slopes downward due to real balance effect.

Step 6: Short-Run vs Long-Run Analysis

■ Short Run:

- SRAS: $P = P_{sr} = 120$ (horizontal line)
- Plug into AD: $Y = 1166\frac{2}{3} + 148,000/120 = 2400$
- Confirms our Step 3 result

■ Long Run:

- LRAS: $Y = \bar{Y} = 2500$ (vertical line)
- Plug into AD: $2500 = 1166\frac{2}{3} + 148,000/P$
- Solve: $P = 148,000/1333\frac{1}{3} = 111$
- Confirms our Step 4 result

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2nframeSummary: Equilibrium Comparison

Variable	Short Run	Long Run
Output (Y)	2400	2500
Price Level (P)	120	111
Real Interest Rate (r)	0.13	0.05
Nominal Interest Rate (i)	0.18	0.10
Consumption (C)	1626	1710
Investment (I)	174	190

Movement from short to long run involves: (1) Output rising to potential, (2) Prices falling, (3) Interest rates declining, (4) Consumption and investment increasing.

2nframeSummary: Equilibrium Comparison

Variable	Short Run	Long Run
Output (Y)	2400	2500
Price Level (P)	120	111
Real Interest Rate (r)	0.13	0.05
Nominal Interest Rate (i)	0.18	0.10
Consumption (C)	1626	1710
Investment (I)	174	190